

Surround Sound Standards

Dolby Digital (AC-3): The AC-3 (Audio Code number 3) surround sound standard was created by Dolby Laboratories. It delivers five audio channels in addition to a Low Frequency Effects (LFE) channel, which is denoted as 5.1 surround sound. Here, the audio information (in a DVD movie for example) consists of a digital stream of encoded information, which needs to be correctly decoded by an external component—in this case, a Dolby Digital decoder—so that you can hear the separate channels of audio. Once the channels have been split up, they are sent to the corresponding speakers to give you a surround sound experience. Technically, Dolby Digital uses six channels, with 18 bits allocated to each at a sampling frequency of 48 KHz. The first movie to feature a Dolby Digital soundtrack was *Batman Returns*.

Dolby Pro Logic: This is an older standard that packs audio information for a centre and a surround channel into your normal stereo channel. This is far easier to implement compared to Dolby Digital. The main difference here is that there is no discrete left and right channel for the surround audio information—the same surround signal goes to both the surround channels (rear speakers).

DTS: An Acronym for Digital Theatre Systems, this is a standard that was formulated by Steven Spielberg and is a competitor to Dolby Digital. It was first debuted in the movie *Jurassic Park*. It uses a sampling frequency of 96 KHz with 24 bits allocated to each of the six audio channels. There are some DVD movies avail-

able with both Dolby Digital as well as DTS soundtracks encoded.

Dolby Digital EX and DTS ES: These are new surround audio standards that add an additional surround sound channel to the 5.1 picture—the rear centre channel. In the case of Dolby Digital EX, this is done by adding a special type of Dolby Pro Logic decoder to the surround channels and creating one more rear centre channel, thus resulting in 6.1 channel sound. A similar strategy is employed by the DTS ES standard where an additional rear centre channel is supplemented to the two surround channels.

THX: An acronym for Tomlinson Holman's eXperiment, THX is often misunderstood as another surround sound technology. Instead, this is a 'seal of approval' standard that involves certification and approval of the components that go into a home theatre such as amplifiers, video playback equipment, speakers, equalizers, etc. THX aims to ensure that the movie-watching audience experiences the highest standard of audio and video in theatres and home audio-video equipment. There are two other sub-standards to the original THX standard—THX Select and THX Ultra. Among other specifications, these were formulated according to the size of the areas the THX system is to be installed in. THX Select is for installations up to 2,000 cubic feet, while THX Ultra is for installations between 2,000 and 3,000 cubic feet.



Mercury SR-301 F, on the other hand, wasn't able to hold its own—its surround sound quality and power output was relatively poor.

MP3 and Audio CD sound quality: Here, we evaluated the ability of the speakers to play back compressed audio (MP3) and uncompressed audio (Audio CD). Among the 2.1 speakers, the Sumoku Alpha - 4R had no problem reproducing mid and high frequency sounds and the audio was pure and sharp across these

Labtec Arena 530

The Labtec Arena 530 is bound to grab your attention with its subtle contoured body. The overall build quality is very good with the satellites featuring grips so that they remain anchored to your desk even at high volumes.

The system provides a function called Matrix 3D (M3D), which can be activated through a button on the sub-

woofer and which simulates four channels of audio from a stereo source such as MP3s or Audio CDs. The connecting cables are fairly long, giving you more freedom in placing the speakers. It's priced at Rs 5,999 and gives impressive audio performance.



Labtec Arena 530						B+
Features	▶	▶	▶	▶	▶	
Performance	▶	▶	▶	▶	▶	
Warranty & support	▶	▶	▶	▶	▶	
Value for money	▶	▶	▶	▶	▶	
OVERALL	▶	▶	▶	▶	▶	